



Cambridge (CIE) IGCSE Computer Science



Your notes

Arrays

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* Arrays



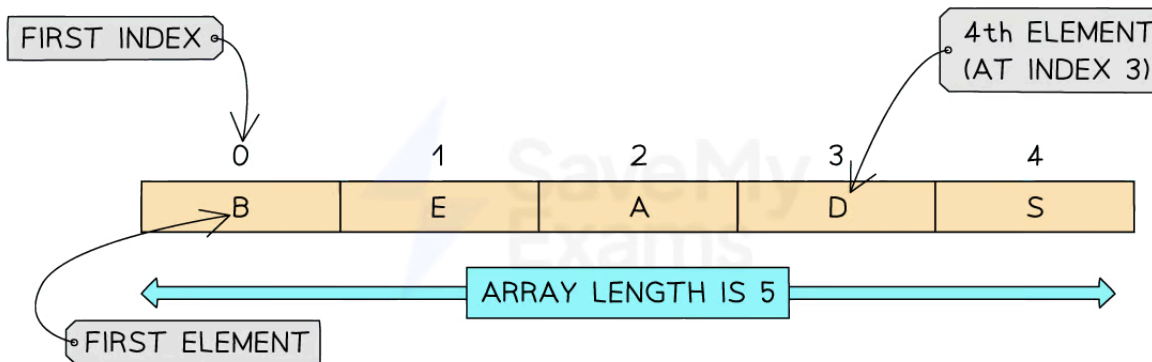
Your notes

Arrays

1-Dimensional Arrays

What is an array?

- An array is an **ordered, static set of elements** in a fixed-size memory location
- An array can only store **1 data type**
- A 1D array is a **linear array**
- Indexes start at generally start at 0, known as **zero-indexed**



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Concept	Pseudocode	Python
Create	DECLARE scores: ARRAY[0:4] OF INTEGER	scores = []
	Creates a blank array with 5 elements (0-4)	Creates a blank array
	scores ← [12, 10, 5, 2, 8]	scores = [12, 10, 5, 2, 8]
	Creates an array called scores with values assigned	
Assignment	colours[4] ← "Red"	colours[4] = "Red"

Assigns the colour "Red" to index 4 (5th element)



Your notes

Example in Python

Creating a one-dimensional array called 'array' which contains 5 integers.

- Create the array with the following syntax:
`array = [1, 2, 3, 4, 5]`
- Access the individual elements of the array by using the following syntax:
`array[index]`
- Modify the individual elements by assigning new values to specific indexes using the following syntax:
`array[index] = newValue`
- Use the len function to determine the length of the array by using the following syntax:
`len(array)`
- In the example the array has been iterated through to output each element within the array. A for loop has been used for this

Python



Your notes

```
# Creating a one-dimensional array
array = [1, 2, 3, 4, 5]

# Accessing elements of the array
print(array[0]) # Output: 1
print(array[2]) # Output: 3

# Modifying elements of the array
array[1] = 10
print(array)    # Output: [1, 10, 3, 4, 5]

# Iterating over the array
for element in array:
    print(element)

# Output:
# 1
# 10
# 3
# 4
# 5

# Length of the array
length = len(array)
print(length)  # Output: 5
```

2-Dimensional Arrays

What is a 2-dimensional array?

- A 2D array **extends** the concept on a 1D array by **adding another dimension**
- A 2D array can be visualised as **a table with rows and columns**
- When navigating through a 2D array you first have to go **down the rows** and then **across the columns** to find a position within the array



Your notes

	0	1	2	3	4
0	B	E	A	D	S
1	S	E	V	E	N
2	W	H	I	T	E

RIGHT INDEX: DETERMINES THE COLUMN

LEFT INDEX: DETERMINES THE ROW

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Concept	Pseudocode	Python
Create	DECLARE NamesAndNumbers : ARRAY[1:3, 1:3] OF STRING	NamesAndNumbers = [],[]
	Creates a blank 2D array with 3 elements (0-2)	Creates a blank 2D array
	NamesAndNumbers[1, 2] ← "John" NamesAndNumbers[1, 3] ← "Emily" NamesAndNumbers[2, 1] ← "25" NamesAndNumbers[2, 2] ← "33" NamesAndNumbers[2, 3] ← "19" NamesAndNumbers[3, 1] ← "Sarah" NamesAndNumbers[3, 2] ← "12" NamesAndNumbers[3, 3] ← "8"	NamesAndNumbers = ["Rob", "Paul", "Hayley"], [10, 5, 8]
	Creates a 2D array called players with values assigned	
Assignment	NamesAndNumbers[0, 1] ← "Holly"	NamesAndNumbers[0][1] = "Holly"

Assigns the name "Holly" to index 0, 1 (1st row, 2nd column) – replaces "Paul"



Your notes

Example in Python

Initialising a 2D array with 3 rows and 3 columns, with the specified values

```
array_2d = [[1, 2, 3],  
            [4, 5, 6],  
            [7, 8, 9]]  
  
# Accessing elements in the 2D array  
print(array_2d[0][0]) # Output: 1  
print(array_2d[1][2]) # Output: 6
```

Iterating through a 2 – dimensions array

- When iterating through an array, a nested for loop can be used

Nested iteration to access items in the 2D array

```
for row in array_2d:  
    for item in row:  
        print(item, end=" ")  
    print() # Print a newline after each row
```



Examiner Tips and Tricks

In the exam, the question will always give an example to demonstrate which order the array is being read from.

Some questions can be X,Y and others can be Y, X. Always refer to the example before giving your answer!



Worked Example

A parent records the length of time being spent watching TV by 4 children



Your notes

Data for one week (Monday to Friday) is stored in a 2D array with the identifier `minsWatched`.

The following table shows the array

		Quinn	Lyla	Harry	Elias
		0	1	2	3
Monday	0	34	67	89	78
Tuesday	1	56	43	45	56
Wednesday	2	122	23	34	45
Thursday	3	13	109	23	90
Friday	4	47	100	167	23

Elias watched 78 minutes of TV on Monday:

- Identify the row for Monday: Row **0**.
- Identify the column for Elias: Column **3**.
- Find the value at `minsWatched[0][3]`: The value is **78**

Write a line of code to output the number of minutes that Lyla watched TV on Tuesday **[1]**

Write a line of code to output the number of minutes that Harry watched TV on Friday **[1]**

Write a line of code to output the number of minutes that Quinn watched TV on Wednesday **[1]**

Answers

- `print(minsWatched[1][1])` OR `print(minsWatched[1,1])`
- `print(minsWatched[4][2])` OR `print(minsWatched[4,2])`
- `print(minsWatched[2][0])` OR `print(minsWatched[2,0])`